

Week 02 Task 4

Linux Basics + Log Files (/var/log)

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Executive Summary

This task is dedicated to develop fundamental linux command line skills which are crucial for cybersecurity professionals as Linux is widely used in penetration testing, security operations, system administration, and digital forensics. During this task, I practiced commonly used Linux commands to gain hands-on experience with file and directory management, system navigation, permissions, process management, system information, and other essential command-line operations.

Scope

This task strengthened my understanding of fundamental Linux commands and improved my ability to work effectively with the Kali Linux terminal. The practical experience gained provides a solid foundation for further cybersecurity-related tasks.such as enumerating skills, reconnaissance, vapt.

Tool Used

Kali Linux

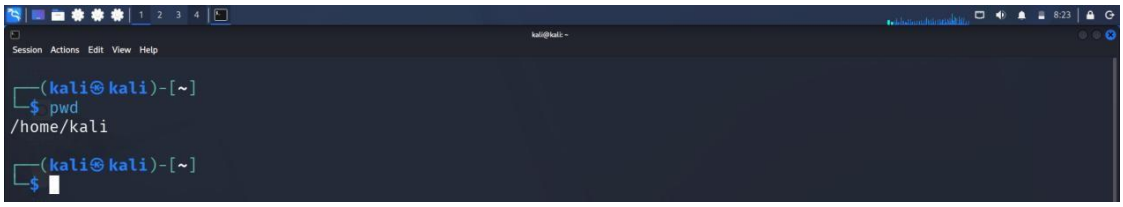
Command 1

Command: pwd

Purpose

pwd command is use to print the current working directory.

Execution of command



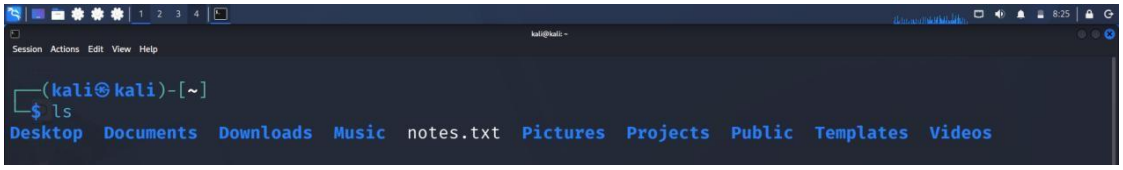
Command 2

Command: ls

Purpose

ls command is use to list all files and folders of current directory.

Execution of command



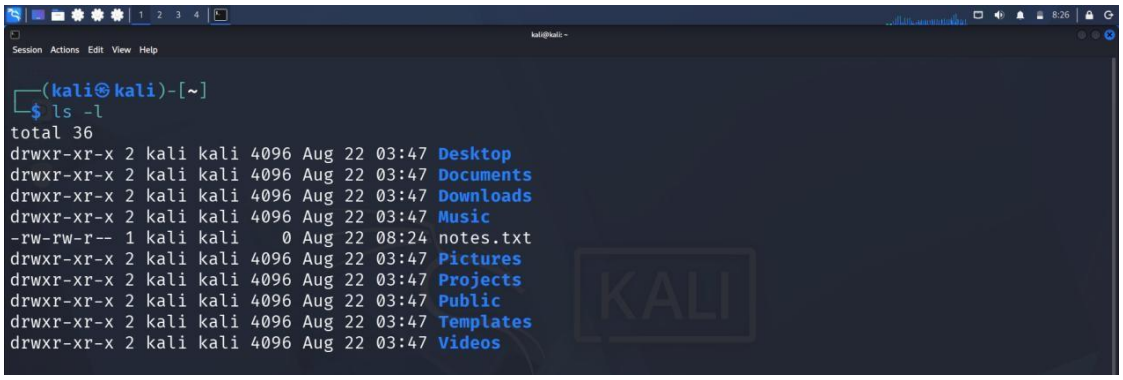
Command 3

Command: ls -l

Purpose

ls -l command is use to list the permission that files and folders have in the current directory.

Execution of command

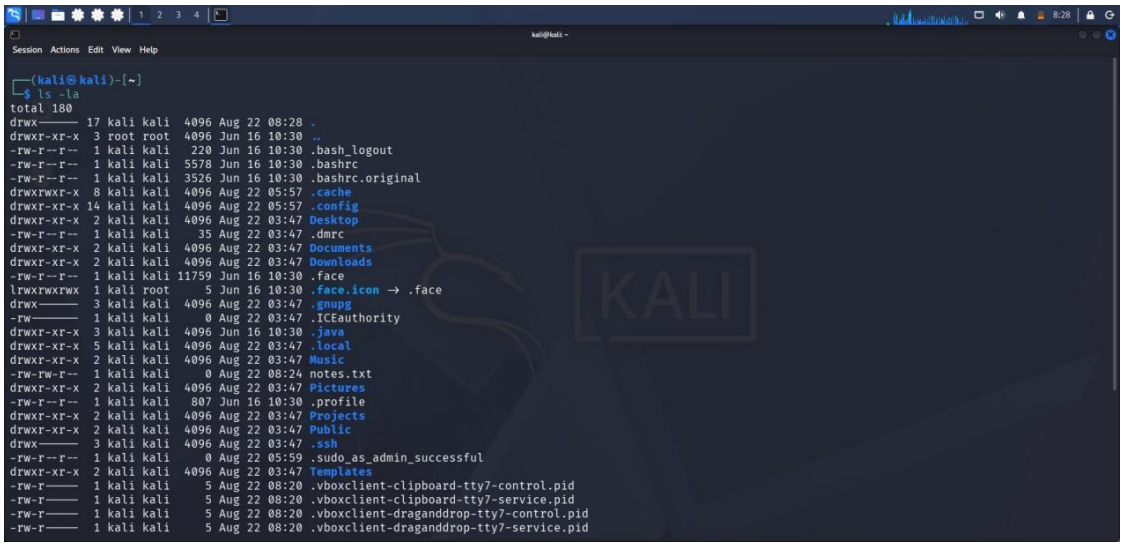


Command 4

Command: ls -la

Purpose: This command is use to list all files and directories in the current folder, including hidden files. Hidden files are identified by a dot (.)

Execution of command

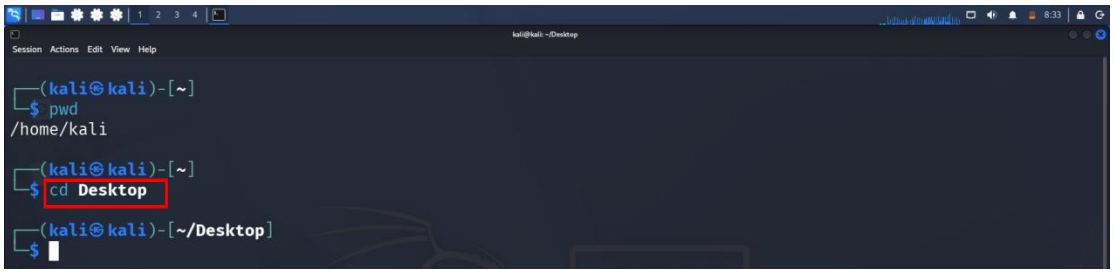


Command 5

Command: cd

Purpose: This command is use to change the directory

Execution of command



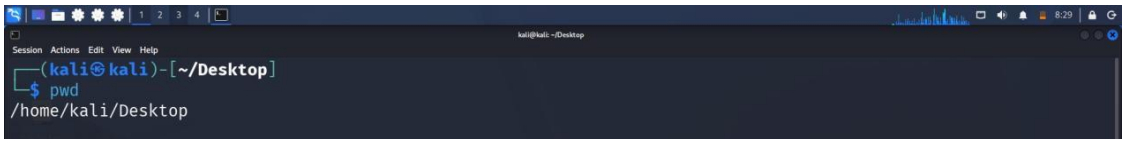
Command 6

Command: cd ..

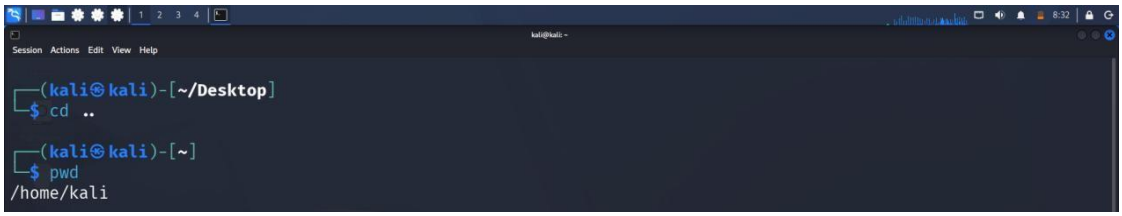
Purpose: This command is use to navigate back one directory level from the current directory.

Execution of command

Currently I am in desktop directory



Go back to /home/kali directory type **cd ..**



Command 7

Command: mkdir

Purpose: This command is use to make a empty directory.

Execution of command



Verification Command : ls



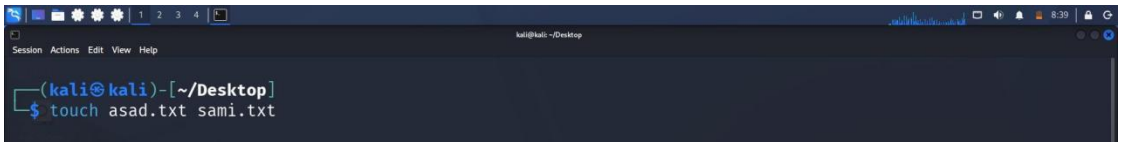
Using **ls** command we have verified folder is created successfully

Command 8

Command: touch

Purpose: This command is use to create empty files

Execution of command



Verification Command: Command is **ls** that verifies files are created using touch command or not



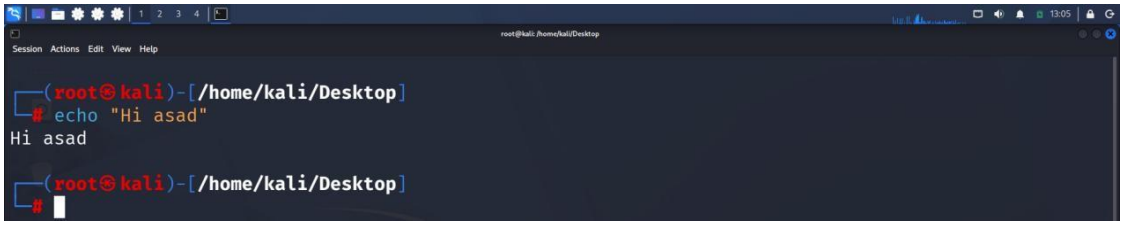
As verified from above Figure both files asad.txt and sami.txt are successfully created using **touch** command.

Command 9

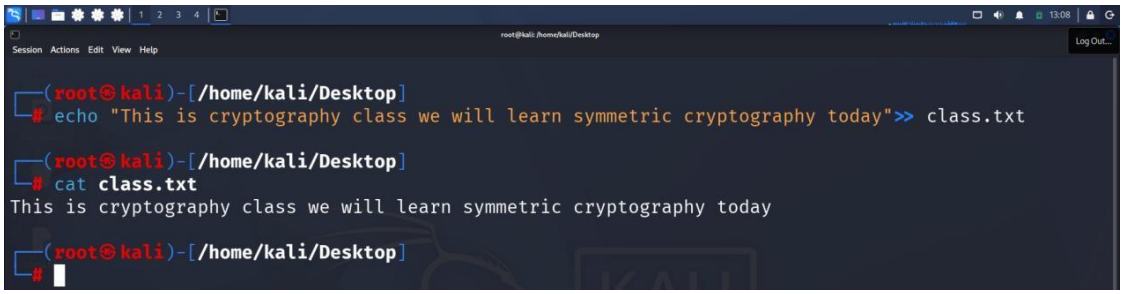
Command: echo

Purpose: This command is use to display the content directly to terminal standard output. It is also use to write content in file

Execution of command



Second example of Echo command



Command 10

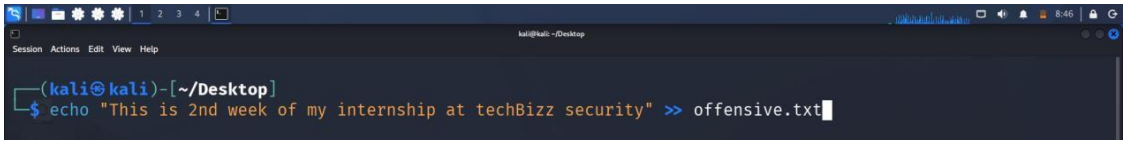
Command: cat

Purpose: This command is use to read the content of file

Execution of command

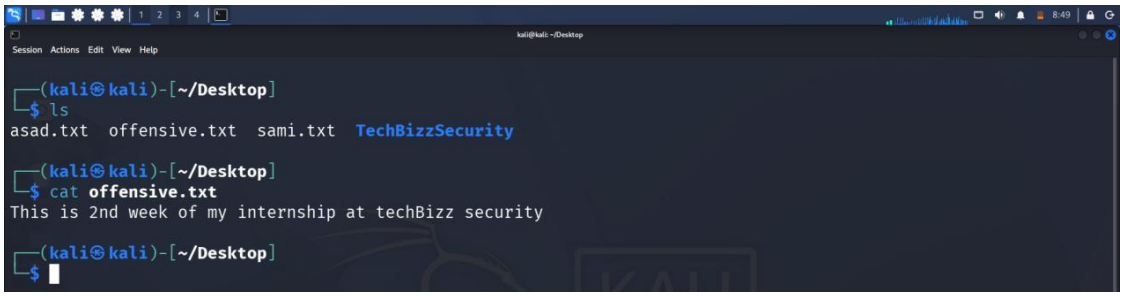
To demonstrate the functionality of cat command that it can read file content first there is a need to create a file with content.

Creating File



Note: File can also be created using nano editor then write some thing in file

Demonstration of cat command

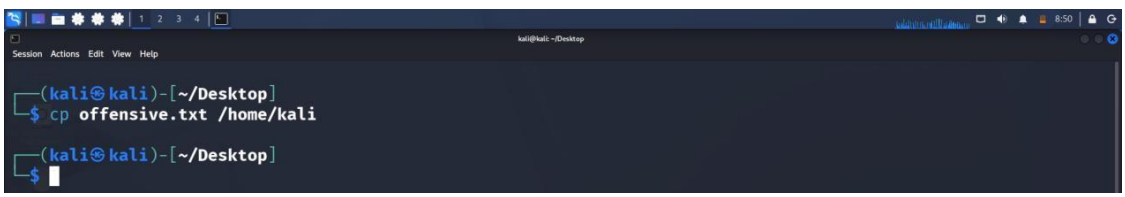


Command 11

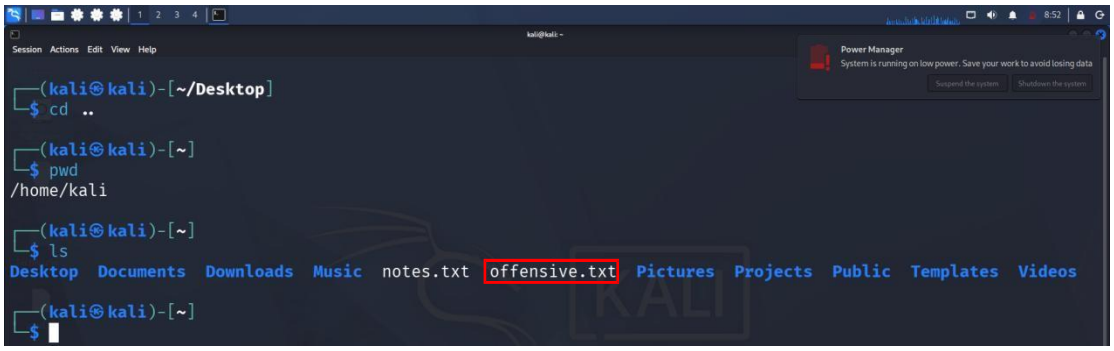
Command: cp

Purpose: This command is use to copy the files and folders

Execution of command



To verify file is copied from desktop to kali folder lets navigate back to kali folder.



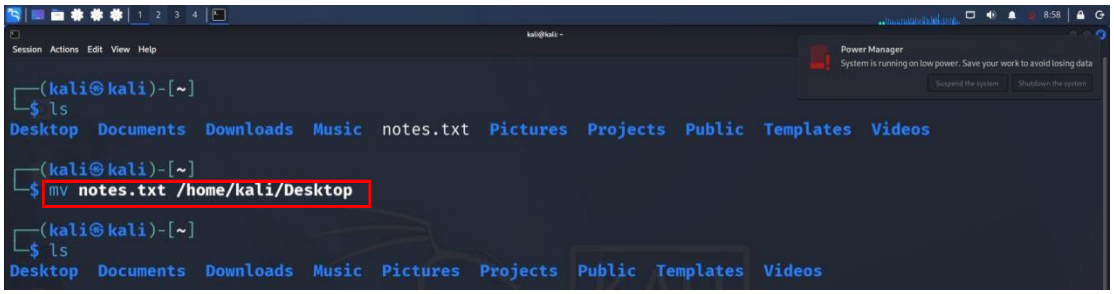
As seen in the above Image file named **offensive.txt** is successfully copied from desktop folder to kali folder.

Command 12

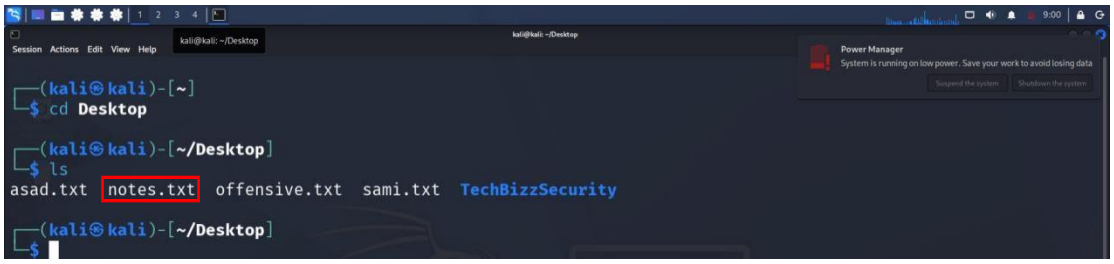
Command: mv

Purpose: This command is use to move files from one folder into another folder It is use to rename a folder.

Execution of command



As File is moved from kali folder to desktop folder to verify navigate to Desktop folder using command **cd Desktop**. Then type command **ls.**



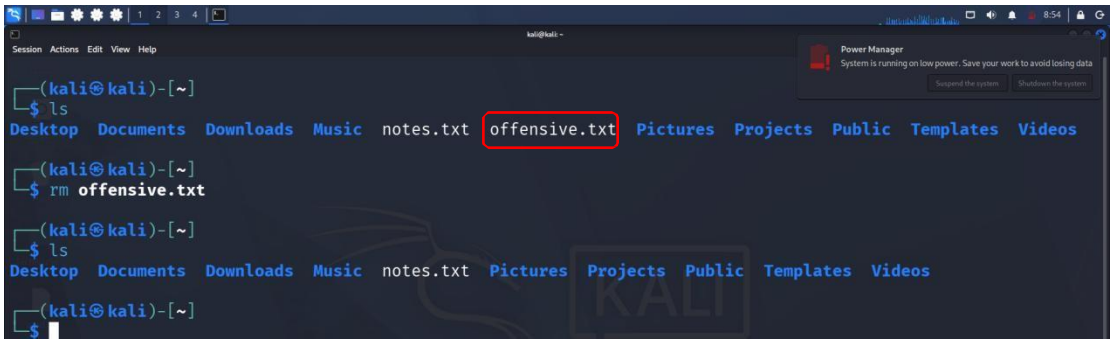
File is successfully moved from kali folder to desktop Folder shown in above Image.

Command 13

Command: rm

Purpose: This command is use to remove file

Execution of command



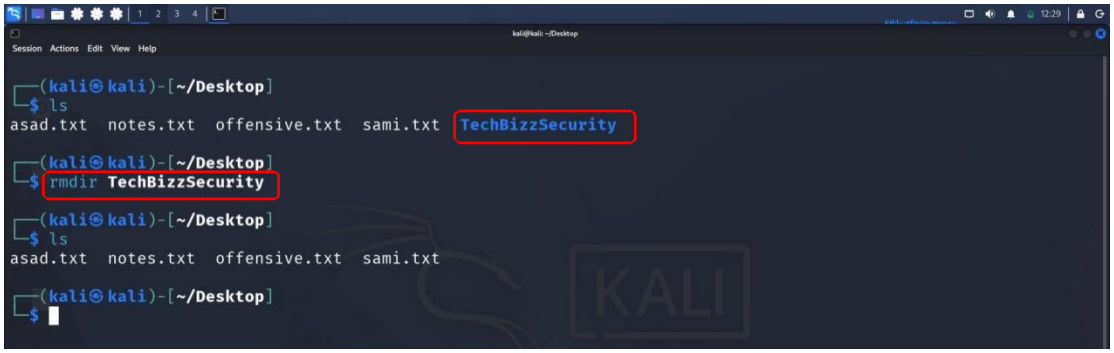
As seen in the above Figure File named **offensive.txt** is successfully removed using **rm** command. In above picture it is confirmed that file is removed using **ls** command after typing rm command.

Command 14

Command: rmdir

Purpose: This command is use to remove the empty folder

Execution of command



rmdir command removes the folder we can check confirm this by typing **ls** in the same working directory as shown in Figure above.

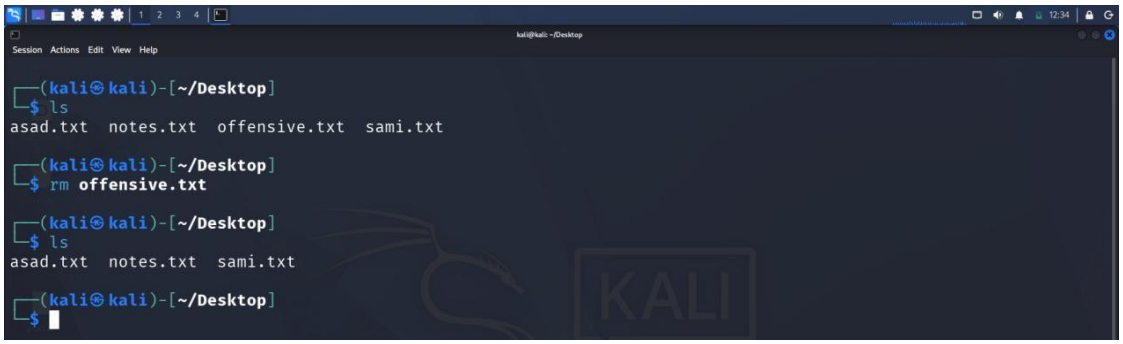
Command 15

Command: clear

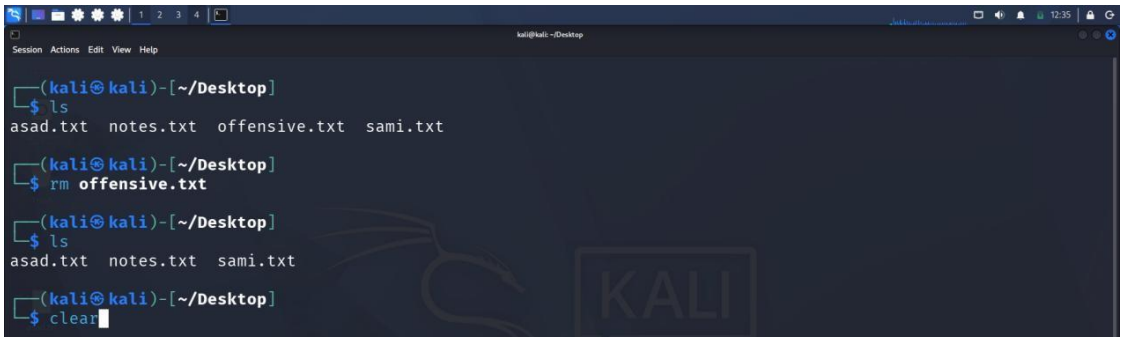
Purpose: This command is use to clear the terminal

Execution of command

Before **clear** command terminal was not clear.



After writing **clear** command.



Terminal is clear after **clear** command.



Command 16

Command: whoami

Purpose: This command is use to print the current logged-in user

Execution of command

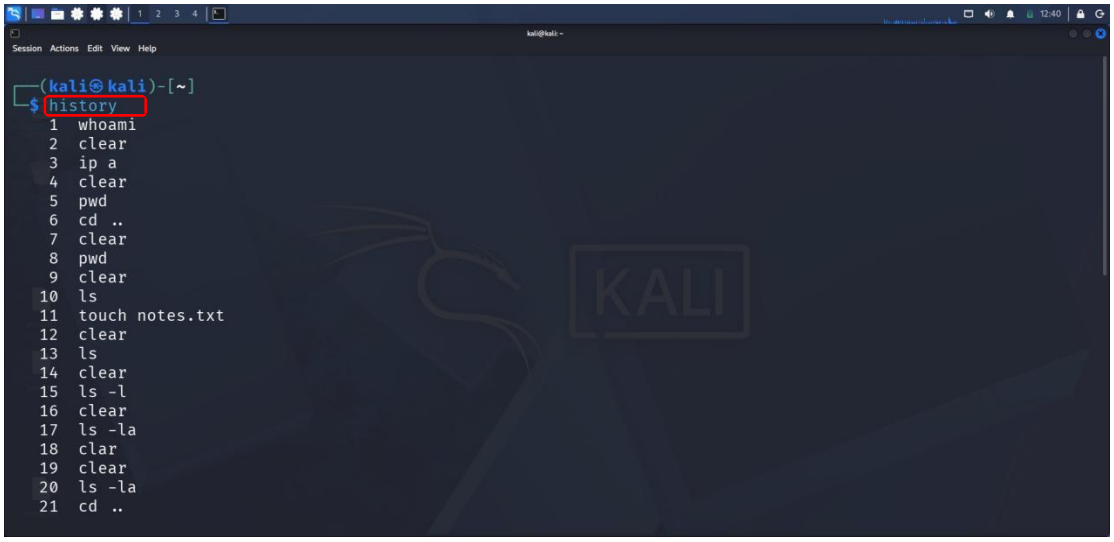


Command 17

Command: history

Purpose: This command is use to check the previous commands used in a terminal

Execution of command



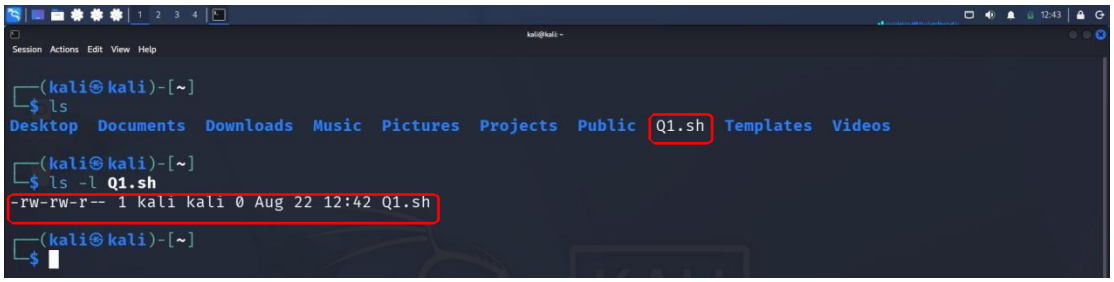
Command 18

Command: chmod

Purpose: This command is use to change the permission of file

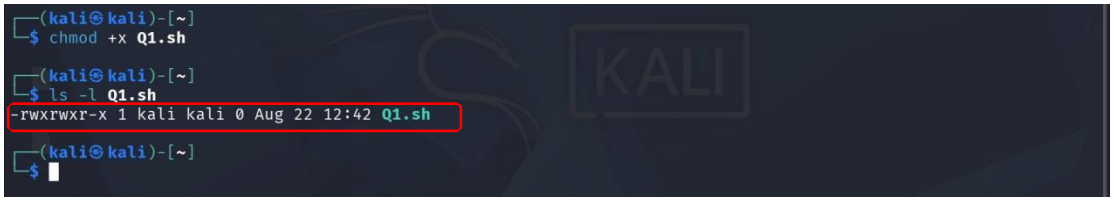
Execution of command

Before Changing the permission of file



As it is clearly seen in the above figure that Q1.sh file has only read and write permission note executable permission.

After Changing the permission of file.



Now File has executable permission also shown in above figure. Chmod +x was executed to upgrade relative permission only and execute privileges without overriding the files existing read/write policies.

Command 19

Command: sudo

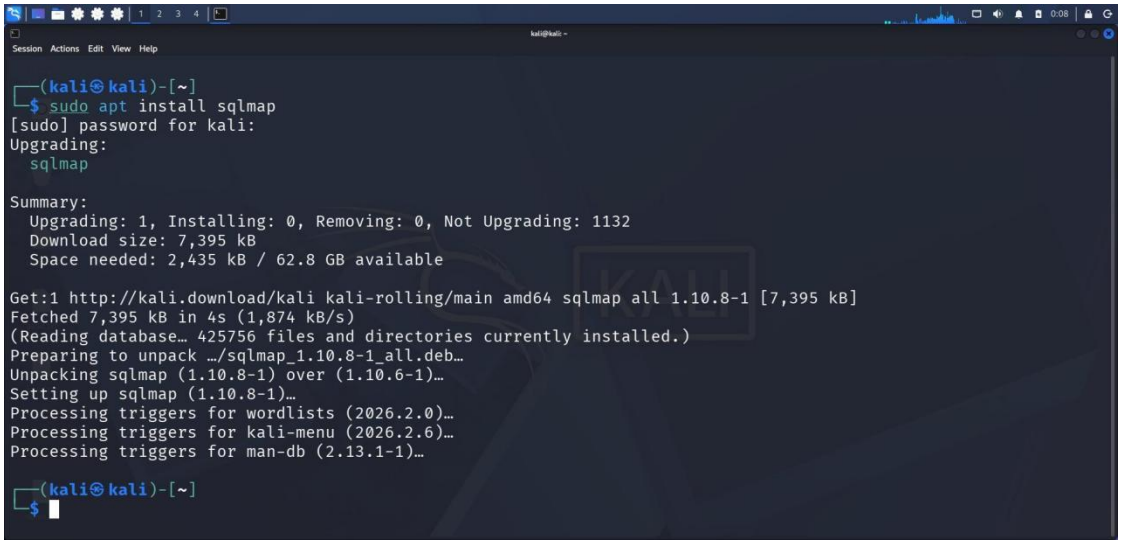
Purpose: This command is use to assign root privileges to user or changing from normal user to root user.

Execution of command

Switching from normal user to root user command **sudo su.**



It is clearly seen in the above image that first user is normal user **kali**. Then user switched from **kali** to **root** by typing command **sudo su**. Whenever we type sudo with any command that means do the work with **admin user** privilges.



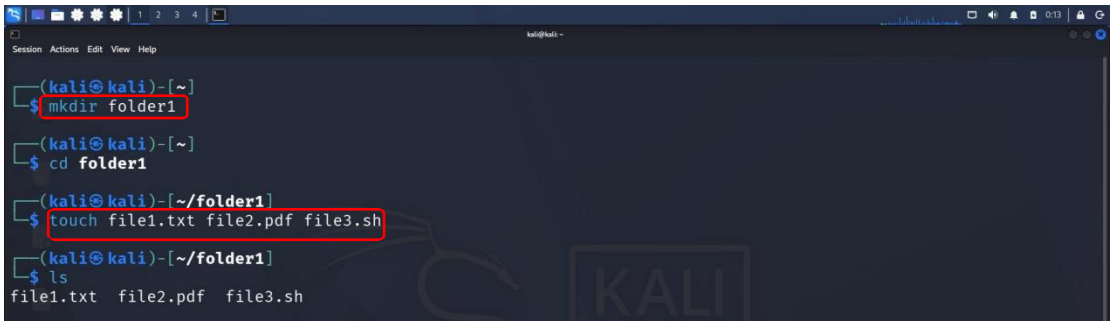
Command 20

Command: rm -r

Purpose: This command is use to delete directory having content means directory having files not empty

Execution of command

First we need to create directory with some files .



In the above figure we have created a folder having three files **file1.txt** , **file2.pdf file3.sh.**



In the above figure it is seen that folder1 is present then we write command **rm -r folder1** then again we check by **ls** command now there is no folder named **folder1**.

Command 21

Command: grep

Purpose: This command is use to search contents in file. It processes text line-by-line and outputs any line that contains a match for the specific text or pattern you are looking for.

Execution of command



Log Files



In the above figure it is shown that I navigated to /var/log directory and find a get a file auth.log that contains all authentication logs.

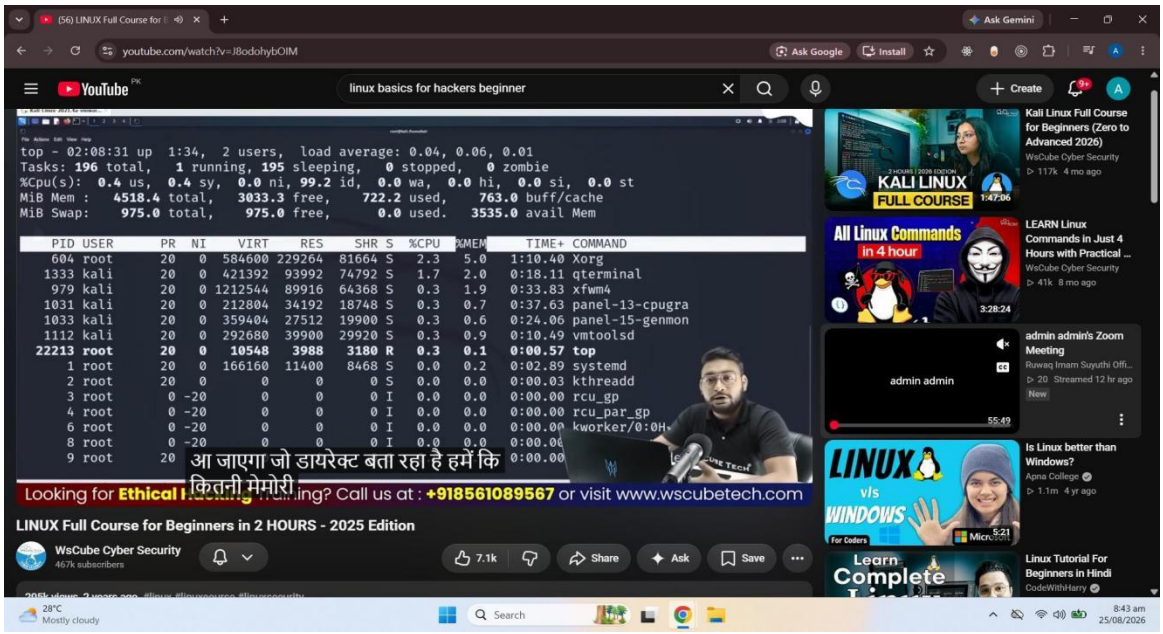


In the above figure it is shown that I grabbed the failed login attempt from auth.log file via grep command.

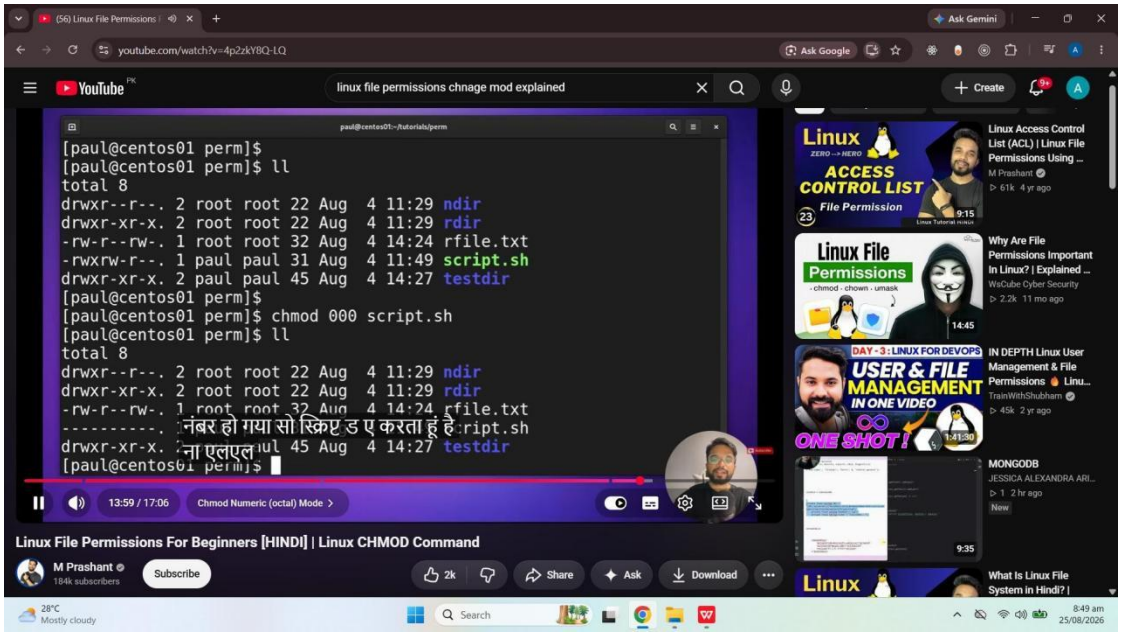
Tryhackme Room Linux Fundamentals



Youtube Video Linux Course



Youtube Video Understanding Linux File Permissions



Conclusion

This task helped me to enhanced my proficiency in linux command line operations and helped me to comfortable with kali linux terminal. This task is a baseline for future cyber security work because linux commands are important in this field.